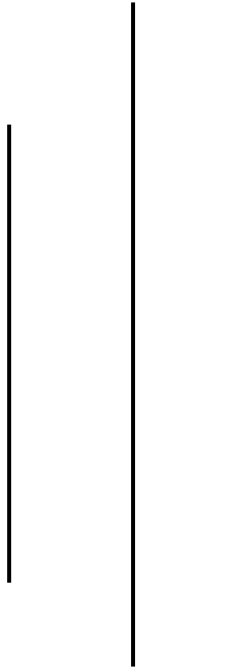


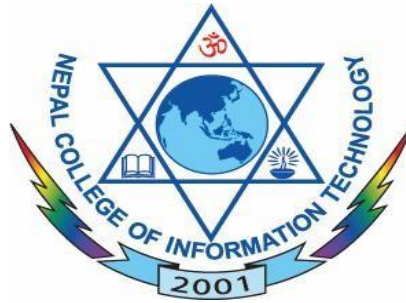
NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

Affiliated to Pokhara University



LAB REPORT FOR MACHINE LEARNING



LAB REPORT 1: ANALYZING AND FORECASTING TIME SERIES TREND USING MOVING AVERAGES

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S. N	LAB TOPIC	DATE	REMARKS
0.	Familiarizing with Numpy, Pandas and Matplotlib	28 th April, 2026	
1.	Analyzing and Forecasting time series trend using Moving Averages	12 th May	

THEORY:

Time Series Data

- A **time series** is data collected over time intervals.
- Used to analyze patterns and predict future values.
- Examples:
 - Yearly enrollment
 - Monthly sales
 - Daily temperature

Components of Time Series

- **Trend** → Long-term increase or decrease
- **Seasonal Variation** → Repeating short-term pattern
- **Cyclical Variation** → Long-period fluctuations
- **Irregular Variation** → Random changes

Moving Average Method

- A forecasting technique used to smooth data fluctuations.
- Helps identify trends in time series data.
- Uses previous observations to predict future values.

Advantages

- Simple to calculate
- Removes random fluctuations
- Useful for short-term forecasting

Simple Moving Average (SMA)

- Average of a fixed number of previous observations.
- Each observation has equal weight.

For 3-Year SMA:

$$SMA = \frac{Y_{t-1} + Y_{t-2} + Y_{t-3}}{3}$$

Features

- Easy and simple method
- Smooths the trend
- Slower response to sudden changes

Weighted Moving Average (WMA)

- Assigns different weights to observations.
- Recent data gets higher importance.

For 3-Year WMA:

$$WMA = \frac{w_1 Y_{t-1} + w_2 Y_{t-2} + w_3 Y_{t-3}}{w_1 + w_2 + w_3}$$

Features

- More accurate than SMA in many cases
- Responds faster to recent changes
- Flexible due to weights

Mean Squared Error (MSE)

- Measures forecasting accuracy.
- Lower MSE means better prediction.

Formula:

$$MSE = \frac{\sum (Y_t - \hat{Y}_t)^2}{n}$$

Where:

- Y_t = Actual value
- $\hat{Y}_t$ = Predicted value
- n = Number of observations

Question: Provide a data table that has all the calculations from moving averages.

- a) Determine the prediction from each moving averages
- b) Compare MSE for each moving averages
- c) Perform visualization for each prediction from moving averages.

Enrollment Data of NCIT College							
Year	Entrollment	3 Year SMA	Error	Error^2	3 Year WMA	WMA Error	Error^2
2015	420						
2016	430						
2017	457						
2018	475	436	39	1547.11	444.00	-31.00	961.00
2019	490	454	36	1296.00	463.43	-26.57	706.04
2020	510	474	36	1296.00	481.00	-29.00	841.00
2021	480	492	-12	136.11	499.29	19.29	371.94
2022	490	493	-3	11.11	490.00	0.00	0.00
2023	494	493	1	0.44	490.00	-4.00	16.00
2024	510	488	22	484.00	490.86	-19.14	366.45
2025	525	498	27	729.00	502.57	-22.43	503.04
2026		510					
	MSE	687.472					

w1	w2	w3
1	2	4

2026 Enrollment Predictions:

Based on 2023, 2024, and 2025 enrollment data:

Method	2026 Prediction (students)	Basis (last 3 years)
3-Year SMA	510	$(494 + 510 + 525) \div 3 = 510.00$
3-Year WMA	512.57	$(1 \times 494 + 2 \times 510 + 3 \times 525) \div 6 = 512.57$

SMA Calculation: $(494 + 510 + 525) \div 3 = 1529 \div 3 = 510.00$ students

WMA Calculation: $(1 \times 494 + 2 \times 510 + 3 \times 525) \div (1+2+3) = (494 + 1020 + 1575) \div 6 = 3089 \div 6 = 514.83$

MSE Comparison

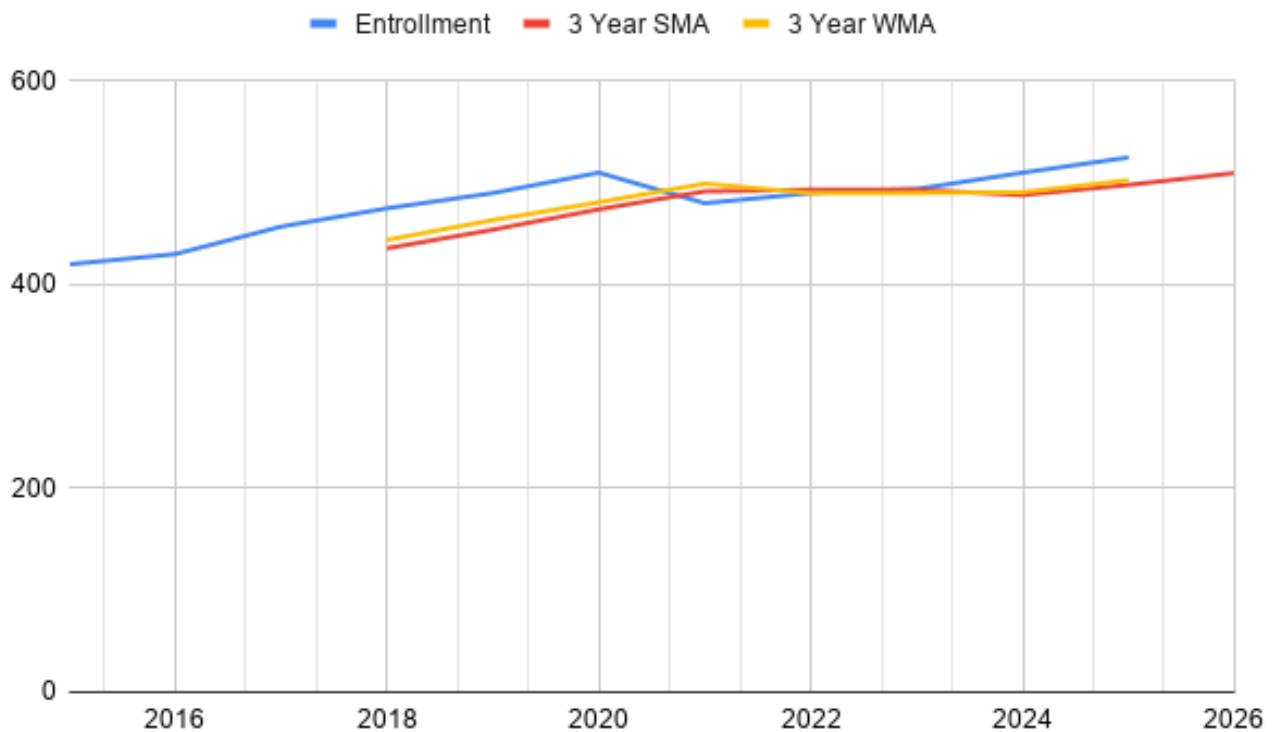
Mean Squared Error (MSE) measures the average of squared forecast errors. A lower MSE indicates better forecasting accuracy.

MSE Formula: $MSE = \Sigma (\text{Actual} - \text{Predicted})^2 \div n$

Method	Sum of Errors ²	n (periods)	MSE	Verdict
3-Year Simple Moving Average (SMA)	5,499.78	8	687.47	
3-Year Weighted Moving Average (WMA)	3,765.47	8	470.68	✓ Better

Conclusion: The 3-Year Weighted Moving Average (WMA) is the superior forecasting method for NCIT enrollment data. Its MSE of 470.68 is approximately 31.7% lower than the SMA MSE of 687.47. This is because WMA assigns greater weight to recent observations, which better captures the underlying upward enrollment trend.

Visualizations:



CONCLUSION:

- Moving averages help smooth time series data and identify trends.
- SMA gives equal importance to all data values.
- WMA gives more importance to recent values.
- MSE helps compare forecasting accuracy.
- From the lab, moving average methods were useful for predicting future enrollment data and analyzing trends effectively.